Tutorial 6 Mini-Lecture: Giant Planets and Small Bodies

The concepts behind Worksheet 6 • Lectures 11–12

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Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

What Worksheet 6 trains

Five problems, all built on Lectures 11–12. The physics you need, per problem:

1. **Giant-planet interiors**: the uniform-density central pressure, the metallic-hydrogen transition, why Saturn needs helium rain.
2. **The Roche limit and rings**: the rigid and fluid limits, why the main rings end where they do, why moonlets survive inside them.
3. **Radioactive dating**: the decay law, short-lived ^{26}Al heating, the long-lived Pb-Pb isochron.
4. **Kirkwood gaps**: the resonance condition, the gap locations, how a resonance empties an orbit.
5. **Cometary activity**: the grey-body temperature, the onset distance, the sublimation rate and its fall-off.

Every problem has the shape and level of one exam question. All parts are analytical; no computer is needed.

Giant-planet interiors and metallic hydrogen

Uniform-density hydrostatic pressure from Lecture 11:

$$\boxed{P_c = \frac{3GM^2}{8\pi R^4}}$$

- Jupiter's bulk density is ~ 1326 $\mathrm{kg\,m^{-3}}$: mostly hydrogen and helium, not rock.
- Metallic hydrogen appears above ~ 100 GPa, where hydrogen becomes an electrical conductor.

Two planets, one transition

Jupiter's central pressure (~ 1200 GPa) is 12 times the metallization value, so its metallic region is large. Saturn's (~ 224 GPa) only just crosses it, so helium separates and rains, supplying Saturn's extra luminosity.

The central pressure decides how much of each planet is metallic hydrogen, and whether helium rain is needed to explain its heat.

The Roche limit and Saturn's rings

The tidal disruption distance of Lecture 11:

$$\boxed{d_R = C\,R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}}$$

- Rigid body: $C = 1.26$; strengthless fluid: $C = 2.46$. The real limit lies between them.
- The fluid limit for ice, ~ 126,350 km, matches the A-ring outer edge to ~ 8%.

Rings inside, moonlets survive

Ice is disrupted out to the fluid limit, so it stays as rings; a denser rocky body survives much closer in. Pan and Daphnis sit between the rigid and fluid limits, held by gravity plus finite strength.

The Roche limit sets where a disk of ice cannot accrete into a moon; the two limits bracket where solid moonlets can still survive.

Dating the Solar System

The decay law and two clocks:

$$\boxed{N = N_0\, e^{-\lambda t}} \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

- ^{26}Al ($t_{1/2}$ = 0.717 Myr) decays in a few Myr: a body forming 2 Myr late has only ~ 14% of the heating.
- Pb-Pb uses two uranium chains; the isochron slope ~ 0.625 at 4.567 Gyr gives an absolute age.

Short-lived vs long-lived

Short-lived isotopes give fine relative timing but are now extinct, so they need an anchor. Long-lived isotopes give the absolute age but blur the first few million years. Together they date the first solids to 4567.3 Myr.

One decay law runs two clocks: ^{26}Al resolves the first few Myr, uranium fixes the absolute age.

Kirkwood gaps and resonances

The resonance location from Kepler's third law:

$$\boxed{a = a_{\mathrm{J}}\left(\frac{k}{j}\right)^{2/3}}$$

- At a $j:k$ resonance the asteroid completes j orbits per k of Jupiter's.
- The $3:1$, $5:2$ and $2:1$ gaps fall at 2.50, 2.82 and 3.28 AU.

Why the gaps are empty

At resonance Jupiter kicks the asteroid at the same phase each cycle; the kicks add, the eccentricity grows, and the orbit becomes planet-crossing. The clearing acts on the semi-major axis, so it shows only in a plot of a, not in a snapshot.

Simple period ratios set the gap positions; coherent resonant kicks pump eccentricity until the body is removed.

Cometary activity and distance

Grey-body balance and sublimation of Lecture 12:

$$T(r) = \left[\frac{S_0(1\,\mathrm{AU}/r)^2}{4\sigma}\right]^{1/4} \qquad Z = \frac{S_0(1\,\mathrm{AU}/r)^2}{4Lm}$$

- $T \approx 278$ K at 1 AU, falling as $r^{-1/2}$; water activity switches on near 3.4 AU.
- Sublimation-limited $Z \approx 4 \times 10^{21}\ \mathrm{m}^{-2}\,\mathrm{s}^{-1}$ at 1 AU, scaling as r^{-2}.

Active only near the Sun

Beyond a few AU the surface is too cold, reradiation takes the energy, and the vapour-pressure cutoff makes production collapse far faster than r^{-2}. Comets are dormant in the outer Solar System.

Temperature sets sublimation; the geometric r^{-2} is only an upper bound, and the true fall-off is steeper.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** ("show that ...") to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Most parts close by asking what the number means: that interpretation is graded, not only the arithmetic.

Worksheet and full solutions: `ips.formingworlds.space/worksheets.html`.
We discuss questions, not presentations of the solutions.